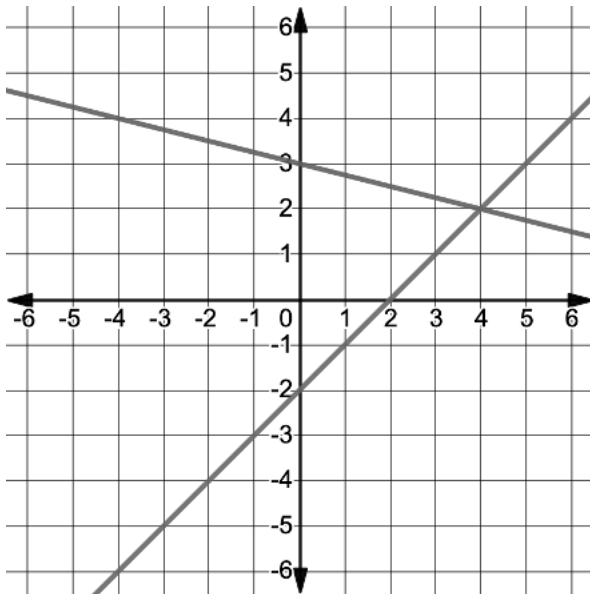


Grade 8
Unit 8
Lesson 7 Practice

Name _____
Date _____
Period _____

The equations $y = -\frac{1}{4}x + 3$ and $y = x - 2$ are shown on the coordinate grid below.



1. Based on the graph, what is the solution to the system of equations $\begin{cases} y = -\frac{1}{4}x + 3 \\ y = x - 2 \end{cases}$?

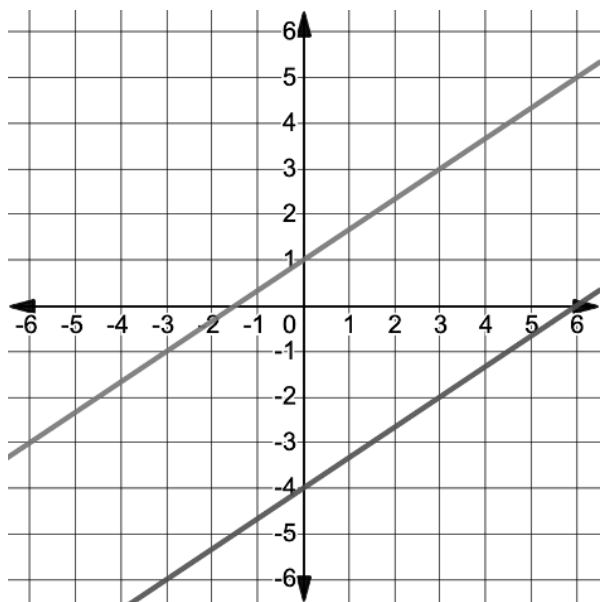
2. Solve the equation $-\frac{1}{4}x + 3 = x - 2$

3. How are your answers to questions 1 and 2 related?

4. Use substitution to verify that your solution from question 1 satisfies both equations $y = -\frac{1}{4}x + 3$ and $y = x - 2$.

$y = -\frac{1}{4}x + 3$	$y = x - 2$

The system of equations $\begin{cases} y = \frac{2}{3}x + 1 \\ y = \frac{2}{3}x - 4 \end{cases}$ is shown on the coordinate grid below.

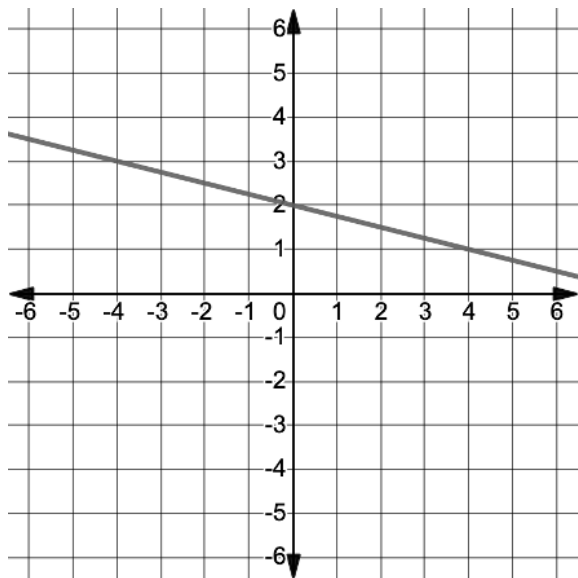


5. Solve the equation $\frac{2}{3}x + 1 = \frac{2}{3}x - 4$.

6. What is the solution to the system of equations $\begin{cases} y = \frac{2}{3}x + 1 \\ y = \frac{2}{3}x - 4 \end{cases}$?

7. Explain how the graph is related to the solution to the equation $\frac{2}{3}x + 1 = \frac{2}{3}x - 4$.

The equation $y = -\frac{1}{4}x + 2$ is shown on the coordinate grid below.



8. Graph the equation $y = -0.25x + 2$.

9. What do you notice about the graphs of the equations $y = -\frac{1}{4}x + 2$ and $y = -0.25x + 2$?

10. Describe the solution(s) to the system of equations $\begin{cases} y = -\frac{1}{4}x + 2 \\ y = -0.25x + 2 \end{cases}$.